Incremental Learning Based on Anchored Class Centers for SAR Automatic Target Recognition

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***Abstract*— Although deep learning methods have achieved great success in synthetic aperture radar automatic target recognition (SAR ATR), their accuracies decline sharply, as new classes are learned, which is known as catastrophic forgetting. The overlapping or confusion between the representations of new and old classes in the feature space is the main cause of catastrophic forgetting. In this article, the incremental class anchor clustering (ICAC) is proposed to address this issue. ICAC solves this problem from three perspectives: 1) how to learn the new classes; 2) how to enable the model to recognize and classify the old classes; and 3) how to solve the imbalance between old classes and new classes. To learn the new classes, ICAC adaptively adds new anchored class centers for new classes, and the features of each new class will be clustered around the corresponding anchored class center. To enable the model to recognize and classify the old classes, ICAC stores some exemplars for the old classes to ensure the classification ability of the old classes without losing the old class centers in the feature space. At the same time, ICAC adopts knowledge distillation to further alleviate catastrophic forgetting. To solve the imbalance between old classes and new classes, ICAC proposes a learning strategy named separable learning (SL), which computes the losses of the old and new exemplars separately and then adds the two losses to make a gradient descent. Experiments on the moving and stationary target acquisition and recognition (MSTAR) dataset and the OpenSARShip dataset demonstrate the effectiveness of this method in SAR ATR.**

***Index Terms*— Catastrophic forgetting, class anchor clustering (CAC), incremental learning, synthetic aperture radar automatic target recognition (SAR ATR).**

I. Introduction

AUTOMATIC target recognition (ATR) of synthetic aperture radar (SAR) can classify the unknown samples into training samples in different SAR target types. In recent years, advances in deep learning have greatly promoted the development of SAR ATR, and the accuracy of SAR ATR has been improved a lot [1], [2], [3], [4], [5]. However, their improvement of accuracy is carried out under the condition of "closed world," which cannot meet the needs of practical applications.

Traditional supervised learning algorithms process the training samples in an offline mode. However, in the real-life applications of SAR ATR, e.g., the recognitions of vehicles, ships, or planes, when a recognition model is training with the existing samples, sensors capture new samples at the same time. The acquisition of sensor data is neither overnight nor one time, but a gradual incremental acquisition process. It requires the recognition algorithm, which can not only train identification model in the case of sufficient samples and classes, but also learn new training samples and new target types in the process of data incremental. In this process, it gradually adapts to the changes of sensor deployment environment and data distribution, so as to obtain stronger identification ability and complete more complex and diversified recognition task. Therefore, for SAR ATR, the real scenario of its application is an incremental acquisition sample and an incremental training sample.

In SAR ATR, there are three basic processes for learning the new samples: 1) recognizing the novel classes; 2) labeling the new training classes; and 3) updating the recognition model. After distinguishing the new and the existent classes, the existing training model needs to be extended to train the samples of new classes [6]. SAR ATR is mostly achieved by designing a classifier, such as support vector machine (SVM) [7] or the neural networks [1], which are both supervised learning algorithms. Unfortunately, for the traditional supervised learning algorithms, the previous recognition abilities of old classes may be lost during sequential training of new classes, which is often called catastrophic forgetting [8]. To enable the model to recognize and classify the old classes as well as the new classes, the model needs to train the entire dataset in an offline mode. In this way, a large amount of storage resources are required to store all the old class samples that have been trained, and the model lacks computational efficiency due to repeated trainings. Thus, it is essential for the SAR ATR models to be incrementally updated with new classes instead of repeated trainings.

On the contrary, in our daily life, unlike the existing machine learning models in SAR ATR, children will memorize the old knowledge when they learn some new skills, rather than completely forgetting them. Incremental learning is a method that can continuously learn new knowledge from new samples, while preserving most of the previously learned knowledge. Compared with offline learning, incremental learning is a more suitable learning mode for many real-world applications. Recent works demonstrate that an incremental SAR ATR system is more efficient and practical [9], [10], [11], [12], [13], [14]. In the field of SAR target recognition, an improved attributed scattering model was introduced in [9]

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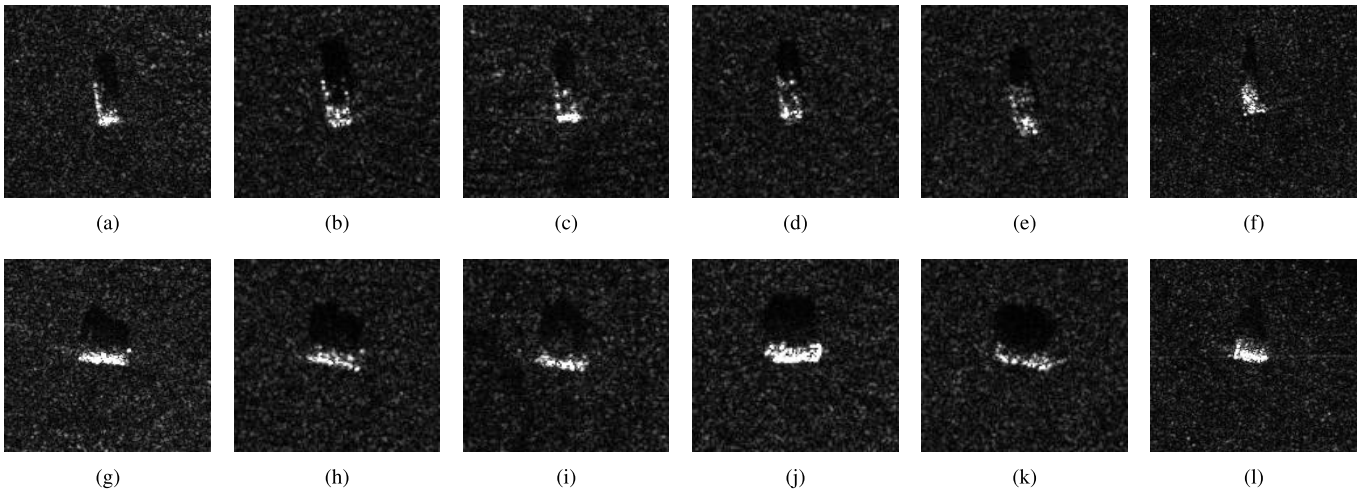


Fig. 1. Six classes of MSTAR samples. (a)–(f) Different types of samples from the same azimuth, which belong to 2S1, BMP2, BRDM2, BTR60, BTR70, and D7, respectively. (g)–(l) Corresponding samples from a different azimuth angle.

to mathematically unify the scattering models of several canonical primitives. A novel incremental Wishart broad learning system (IWBLS) was designed in [10] to achieve polarimetric SAR (PolSAR) image classification. An incremental kernel nonnegative matrix factorization (IKNMF) was shown in [11] to reduce the computing requirements for large-scale data in hyperspectral unmixing. An incremental nonnegative matrix factorization (INMF) method was proposed in [13] for SAR target recognition to overcome the defect of conventional methods in online processing. An open set model with incremental learning (OSmIL) method was introduced in [14], which can learn the new unknown classes continually. In the field of SAR target detection, a new remote sensing (RS) incremental object detection architecture was presented in [12], which is based on feature pyramid and knowledge distillation.

There is another serious problem in SAR ATR. Due to the SAR imaging mechanism, the differences of intraclass samples are relatively large, but for interclass, the result is opposite. The reason for the large intraclass difference is that the various azimuth angles within one class vary greatly. For different classes of objectives from the same angle, they are basically the same, which is the cause of the latter. Fig. 1(a)–(f) represents different types of samples from the same azimuth. Fig. 1(g)–(i) represents the corresponding samples from different azimuth angles. From Fig. 1(a)–(f), it can be seen that the characteristics of different types of samples from the same azimuth are similar except Fig. 1(f). However, the samples from different azimuth angles of the same class, such as Fig. 1(a) and (g), have big differences in characteristic.

To prove that, a t-distributed stochastic neighbor embedding (t-SNE) [15] visualization method is used for low-dimensional visualization for six categories of samples in moving and stationary target acquisition and recognition (MSTAR) dataset, as shown in Fig. 2. The t-SNE is an embedding model that can map data in high-dimensional space to low-dimensional space and retain the local characteristics of the dataset. After t-SNE transformation, if it is separable in the low-dimensional space, the data are separable; otherwise, the data are not separable.

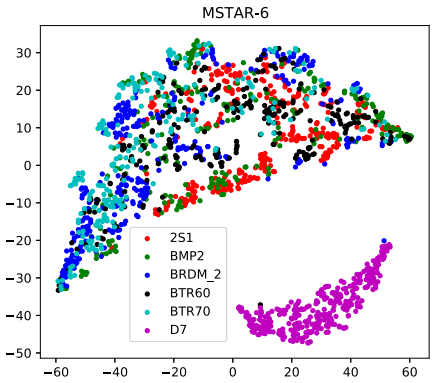


Fig. 2. Result of the t-SNE dimensionality reduction visualization of six classes in Fig. 1. These five categories' features (2S1, BMP2, BRDM2, BTR60, and BTR70) are extremely similar, and the features of the same class are relatively diverged, which are visually the same as Fig. 1 from the final effect.

As can be seen in Fig. 2, the sample characteristics of five classes, 2S1, BMP2, BRDM2, BTR60, and BTR70, are very similar, and the distances between classes are small. At the same time, the feature distribution of each class is relatively scattered with a larger intraclass distance. It is consistent with the effect we see visually in Fig. 1.

As the number of classes is equal, traditional supervised recognition can get a model that works well for all classes. Nevertheless, in the process of class incremental recognition, the class samples come as a stream, and the original distribution of old class samples will be influenced by the arrival of new class samples deeply. The degree of the influence depends on two aspects; one is the size of new class samples to the old class samples, and the other is the similarity between the features of the new classes and the old classes. The reason why we did this is that we wanted to show that most of the classes in the MSTAR dataset are very similar. If each class can be separated spatially, the features differences between different classes will be more obvious, which will facilitate incremental recognition on the MSTAR dataset.

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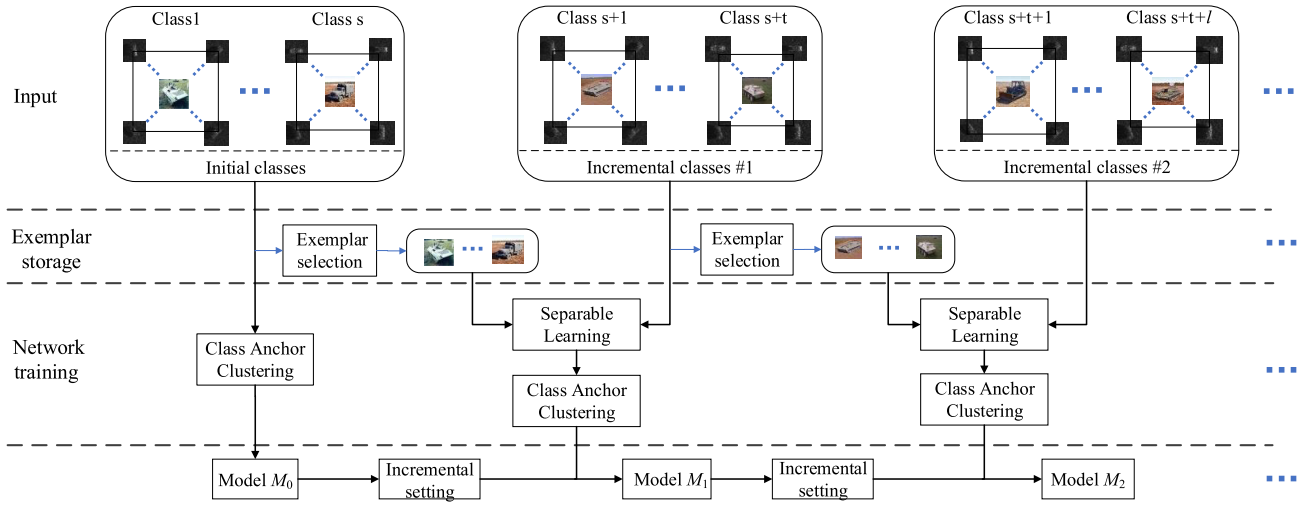


Fig. 3. Incremental learning scenario of ICAC. The model M_0 that can classify s classes is created by training the initial s classes with CAC. After training, the same number of exemplars is stored for each class. When t new classes arrive, for the new classes, t new output nodes and anchored class centers will be generated in network to adapt to these correspondence classes, which is the definition of incremental setting. The s old exemplars will be trained together with the t new classes samples by SL and CAC. After that, model M_0 is extended to M_1, and the same number of exemplars is stored for the new t classes.

In the process of incremental learning for SAR target classes, because the SAR target is characterized by large intraclass distance and small interclass distance, the distributions of new class and old class are prone to aliasing, which leads to the error discrimination of the model. Therefore, in this article, the incremental class anchor clustering (ICAC) method is proposed for SAR incremental object recognition, as shown in Fig. 3. The novelty of ICAC relative to the previous methods is mainly reflected in the following two aspects. First, ICAC adaptively adds new anchored class centers to learn the new classes, so as to carry out effective incremental learning of new classes. Second, ICAC utilizes separable learning (SL) to solve the class imbalance between old classes and new classes. For the SAR images' characteristics of large intraclass distances and small interclass distances, ICAC utilized class anchor cluster (CAC) to effectively solve this problem by explicitly gathering the samples of each class around the corresponding anchor in the network training stage to reduce the intraclass difference, at the same time, making different types of samples separate as far as possible to increase the interclass difference. ICAC sets a fixed number of anchored class centers for the initial known classes and clusters each known class around the corresponding anchored class center. In other words, the known classes are intraclass compact and interclass separable in the feature space.

When the new classes come, ICAC proposes to cluster the new class by adaptively adding new anchored class centers. Meanwhile, ICAC will reserve part of the old class samples, which called exemplars and cluster them around the previous corresponding anchored class centers to ensure the classification ability of the old classes without losing the old class centers. In this way, the representations of all the classes in the feature space are nonoverlapping, which can classify the unknown samples by the nearest-mean-of-exemplars rule. Experimental results on the MSTAR dataset and the OpenSARShip dataset demonstrate that the proposed ICAC outperforms the state of the arts on the accuracy of class-incremental recognition.

The rest of this article is organized as follows. In Section II, the related work of this method is introduced. In Section III, the details of ICAC are given. The experiments and result analysis are introduced in Section IV. Finally, Section V introduces the conclusions of the proposed method.

II. Related Work

ICAC is an incremental learning algorithm based on CAC. The latter is used to increase the interclass distance and reduce the intraclass distance, which is a metric learning method. Therefore, the related work includes two sections: metric learning and incremental learning.

A. Metric Learning

Metric learning is a branch of machine learning that aims to learn distances from the data, which enhances the performance of similarity-based algorithms [16]. In the feature space, all data are converted into feature vectors, and the distances between the feature vectors of similar data are small, while those of dissimilar data are large, so as to distinguish these data. Metric learning is applied in many fields, such as image retrieval, face recognition, target tracking, and so on.

Distance loss functions are applied in metric learning to learn meaningful feature embeddings. Contrastive loss [17] was raised to make the distance between positive sample pairs as small as possible, and the negative part was just the reverse. Triplet loss [18] was introduced to learn representations in a compact Euclidean space where distances were directly matched to a measure of similarity. Then, an extension of triplet loss named tuplet loss was proposed in [19], which maximized the distances from one input to multiple negative samples rather than only one. CAC loss was proposed in [20], which modified the tuplet loss and used the anchored class

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centers to form tight and class-specific clusters for inputs. Triplet center loss in [21], as the term suggests, was the combination of triplet loss and center loss, which mainly concentrated on the center of samples when calculating the distances.

In the field of RS, many works focus on the relative and absolute position of data in space to realize the feature learning of data space. A new deep metric learning approach was proposed in [22] for RS images, which improved scene discrimination capability through two different components. A scalable neighborhood component analysis (SNCA) aimed at constructing the neighborhood structure in the metric space, and the cross entropy (CE) loss aimed at preserving the class discrimination capability. Distribution shift metric learning (DML-ds) was introduced in [23], which improved the original Laplacian-regularized metric learning by adding an interclass distribution shift regularization term. A deep metric learning approach with generative adversarial network regularization (DML-GANR) was developed in [24] for high-spatial-resolution RS images (HSR-RSI) retrieval. A spatial metric learning method was proposed in [25] to obtain a difference image that is more robust to the speckle by learning a metric from a set of constraint pairs. A two-stage training method was introduced in [26], which uses CNN and metric learning for SAR ATR.

B. Incremental Learning

Recently, incremental learning gains wide attention due to the rise of deep learning algorithms [27], [28], [29]. According to the related work [29], class-incremental learning (class-IL) is supposed to satisfy the following conditions.

1) The model is able to train from data stream, with classes appearing in any order and at any time.
2) A multiclass classifier should be provided by the model for the existed classes at any time and has a good performance on classifying both old and new classes.
3) The model is supposed to contain a reasonable number of parameters and memory requirements according to the number of classes seen so far.

Conditions 1 and 2 reveal the essence of class-IL, and condition 3 prevents such algorithms that store all training examples and retrain the model when new classes arrive. There are three crucial components in class-IL: a memory buffer to store few exemplars from old classes, a forgetting constraint to keep previous knowledge, and a learning system that balances old and new classes [27].

However, saving exemplars will consume storage resources at the same time. Various works [27], [29] have adopted knowledge distillation [30] in class-IL to restore previous knowledge. Later, new forgetting constraints have been proposed in [31], which presents the usage of feature-level distillation by penalizing the change of feature representation from the old model. However, there is still a long way to totally get rid of catastrophic forgetting.

Moreover, another serious problem comes into sight: the imbalance between past and new classes creates a bias in the class weight vectors in the last linear layer. When learning new classes, the distribution of the new class samples is more complete than that of the old class samples because of the larger number of samples in the new class, and the network prediction will be biased to new classes. To deal with it, nearest-class-mean (NCM) was impactful to counter this imbalance in [29]. Hou *et al.* [31] rectified the bias by cosine normalization in the last layer.

Some works [14], [32], [33], [34], [35], [36] extended incremental learning and combined incremental learning with open-set recognition. In this way, the model can not only recognize the new unknown classes, but also learn the new classes without retraining. Open world recognition (OWR) [35] formalized the problem of OWR and provided an open world evaluation protocol. An extreme value machine (EVM) [36] proposed a "one versus the rest"-based multiclass open-set classifier, which contains a well-grounded interpretation derived from the statistical extreme value theory (EVT). Then, OSmIL [14] extended EVM by selecting edge samples and demonstrated the effectiveness in SAR ATR. A new open world learning (OWL) method was proposed based on meta-learning in [32], which maintained only a dynamic set of seen classes, allowing new classes to be added or deleted without model retraining. The OWR problem was addressed in [34], and three extensions were proposed to its current formulation. A probabilistic approach was introduced in [33] to unify open-set recognition with the prevention of catastrophic forgetting in deep continual learning. The works mentioned above are more in line with practical applications.

III. Incremental CAC

A. Class Anchor Clustering

CAC [20] is a loss that prompts the known training samples to form tight clusters around anchored and class-specific center points in the feature space. It is the first method that trains with a distance-based loss when using distance during the test. By definition, the class-specific centers are anchored class centers for known classes in the feature space. Each known class i corresponds to an anchored class center $\boldsymbol{c}_i$ in the feature space. After passing through the network, n known classes will eventually map to n-dimensional feature space. The anchored class center for each known class is a point on its class coordinate axis, which is equivalent to a scaled standard basis vector $\boldsymbol{e}_i$ or a scaled one-hot vector. The definition of the anchored class center is as follows:

$$\boldsymbol{C} = (\boldsymbol{c}_1, \ldots, \boldsymbol{c}_N) = (\alpha \cdot \boldsymbol{e}_1, \ldots, \alpha \cdot \boldsymbol{e}_N) \tag{1}$$

$$\boldsymbol{e}_1 = (1, 0, \ldots, 0)^T, \quad \boldsymbol{e}_N = (0, \ldots, 0, 1)^T \tag{2}$$

where α is the magnitude of the anchored class center. The samples for each class will be clustered toward their corresponding anchored class center during training. After training, the anchored class center positions $\boldsymbol{C}$ are adjusted to the mean position of the correctly classified training data.

For the CAC loss, it is a distance-based loss that encourages training inputs to minimize the intraclass distance and maximize the interclass distance between different classes. On the one hand, to minimize the intraclass distance, the distance between the training logits $f(\boldsymbol{x})$ and the corresponding anchor

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class point $\boldsymbol{c}_y$ is supposed to be small; then, the anchor loss term is defined as follows:

$$\mathcal{L}_A(\boldsymbol{x}, y) = d_y = \left\| f(\boldsymbol{x}) - \boldsymbol{c}_y \right\|_2. \tag{3}$$

On the other hand, to maximize the distance to the anchor point $\boldsymbol{c}_y$ and all other anchor points, the definition of discriminative loss (DL) is as the following:

$$\mathcal{L}_D(\boldsymbol{x}, y) = \log\left(1 + \sum_{j \neq y}^{N} e^{d_y - d_j}\right). \tag{4}$$

By uniting the anchor loss and DL, then the CAC loss is defined as follows:

$$\mathcal{L}_{\text{CAC}}(\boldsymbol{x}, y) = \mathcal{L}_D(\boldsymbol{x}, y) + \lambda \cdot \mathcal{L}_A(\boldsymbol{x}, y) \tag{5}$$

where λ is a hyperparameter that balances these two individual loss terms. Using the CAC loss, the distance between the training input and the ground-truth anchored class center is minimized, while the other centers keep maximized distances from inputs.

B. Incremental Learning

The aim of class-IL is to learn a unified classifier in a sequential data stream from different classes, i.e., the sample sets $(X_1, X_2, \ldots, X_{s+t})$, where X_y contains all images from class y. Processing a batch of classes can be seen as a task T. For the new task T_i, the data contain the whole samples of the new set of classes $\mathcal{X}$ $(X_{s+1}, \ldots, X_{s+t})$ and a small quantity of exemplar data $\mathcal{P}$ $(P_1, \ldots, P_s)$ from previous classes $(X_1, \ldots, X_s)$, and the model should classify all the existed classes.

The network model Θ consists of a feature extractor $\varphi(\cdot)$ and a fully connected layer $fc(\cdot)$. Different from the traditional multiclass classifier, which depends on the softmax function, ICAC uses a nearest-mean-of-exemplars classification strategy, which is inspired by incremental classifier and representation learning (iCaRL). A reasonable number of exemplars are selected in a certain way to provide an approximation to the class mean.

There are two main differences between ICAC and iCaRL when calculating the prototype vector for each class. First, iCaRL computes the prototypes after the $\varphi(\cdot)$ layer, while ICAC computes the prototypes after the $fc(\cdot)$ layer. Second, the total number of exemplars saved by iCaRL for old classes is fixed. Therefore, when there are fewer old classes, the number of exemplars retained by each old class will be larger, and the situation will be opposite for the more old classes. However, ICAC maintains a fixed number of exemplars for each old class, so the total number of exemplars preserved for older classes increases, as the number of older classes grows.

In the training stage, first, a multiclassifier is trained, i.e., $y \in y[1:s]$, according to the initial task T_0. After the data from s classes pass through the network, an s-dimensional output vector o is generated, and s anchored class centers are initialized for the s classes. As described in Section III-A, the corresponding output vector o of each class will finally cluster around the corresponding anchored class center. After training T_0, to ensure the classification ability to the old classes and not to lose the old class centers, the exemplar set is chosen, and the prototype vector is computed for each training class. A growing size of memory along with the class number is allowed instead of remaining stationary (K), but the quantity of chosen samples keeps fixed (m) for each training class as exemplars to prevent catastrophic forgetting. Since ICAC encourages the clustering behavior during training, the distances between the feature vectors of each class are small. For the sake of simplicity, the exemplars for old classes are selected randomly. The advantage of randomly selecting exemplars is that the original distribution of classes can be randomly covered, thus ensuring the diversity of selected exemplars.

Second, when T_1 arrives, for the new classes, the $fc(\cdot)$ layer is adapted to learn new classes $(y \in y[s:s+t])$ by adding t new output nodes; correspondingly, t new anchored class centers are added for t new classes. Then, the samples from t new classes will eventually cluster around the corresponding t new anchored class centers after training.

To alleviate catastrophic forgetting, knowledge distillation is applied in this model as the forgetting constraint, which penalizes the change concerning the output of the old model Θ_s using Kullback-Leibler (KL) divergence to preserve the network's knowledge about the old classes [30]. The distillation loss $(\mathcal{L}^{\text{KD}})$ for the exemplar sets $\mathcal{P}$ and the samples from the new classes are computed.

Different from joint training, the training process of incremental learning is carried out under the condition of class imbalance. The main reason is that incremental learning can only retain a small number of old class exemplars, for example, 30 exemplars per old class. When the new class comes, the number of new class samples is often much larger than 30, so there is an imbalance in the number of training samples between the old and new classes. Since the training of the model is obtained by fitting the current training data, the unbalanced training set causes the model to tend to identify the classes with larger sample size.

To solve the imbalance between old classes and new classes, a learning strategy named SL is proposed, which is inspired by the compositional learning system (CLS) in [27]. SL computes two separate losses for $\mathcal{X}$ and $\mathcal{P}$ mini-batches and adds the two losses to make a gradient descent

$$\mathcal{L}_{\mathcal{X}} = \mathcal{L}_{\mathcal{X}}^{\text{CAC}} + \xi * \mathcal{L}_{\mathcal{X}}^{\text{KD}} \tag{6}$$

$$\mathcal{L}_{\mathcal{P}} = \mathcal{L}_{\mathcal{P}}^{\text{CAC}} + \xi * \mathcal{L}_{\mathcal{P}}^{\text{KD}}. \tag{7}$$

The adaptive weighting function ξ between the classification loss and the distillation loss is defined as the following:

$$\xi = \xi_{\text{base}} \left(\frac{C_n + C_0}{C_n}\right)^{2/3} \tag{8}$$

where C_n denotes the number of new classes, and C_0 denotes the number of old classes. ξ_{base} is 5.

The classification loss for the new classes $(\mathcal{L}_{\mathcal{X}}^{\text{CAC}})$ and the old classes $(\mathcal{L}_{\mathcal{P}}^{\text{CAC}})$ is computed by CAC loss, which is defined in (5). CLS uses o_{new} of new classes to compute the CE loss. Different from CLS, ICAC utilizes $o = \{o_{\text{old}}, o_{\text{new}}\}$ of new

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classes to compute the CAC loss. Additionally, the exemplar set is complied after learning the incremental task in this situation, but it is before the task in CLS. So, $\mathcal{L}_{\mathcal{P}}^{\mathrm{KD}}$ contains only the distillation losses of the old class exemplars without the new. Distillation loss ($\mathcal{L}_{\mathcal{X}}^{\mathrm{KD}}$) is computed using o_{old} and the output of the previous model $\hat{\Theta}^s(x)$

$$\mathcal{L}_{\mathcal{X}}^{\mathrm{KD}} = D_{\mathrm{KL}}\left(\sigma\left(\hat{\Theta}^s(x)\right) \| \sigma\left(o_{\mathrm{old}}\right)\right). \tag{9}$$

For the distillation loss ($\mathcal{L}_{\mathcal{P}}^{\mathrm{KD}}$), an exemplar set P is compiled, which contains the fixed number of exemplars from the old classes. For $(x', y') \epsilon P$, distillation loss ($\mathcal{L}_{\mathcal{P}}^{\mathrm{KD}}$) is computed similar to (9)

$$\mathcal{L}_{\mathcal{P}}^{\mathrm{KD}} = D_{\mathrm{KL}}\left(\sigma\left(\hat{\Theta}^s\left(x'\right)\right) \| \sigma\left(o'_{\mathrm{old}}\right)\right) \tag{10}$$

where $o' = \{o'_{\mathrm{old}}, o'_{\mathrm{new}}\}$ is the output vector of exemplar set $\mathcal{P}$. Finally, a gradient descent is achieved in SL by adding the two losses.

In SL, the network can simultaneously learn both old exemplars and new class samples, while making a gradient descent. In this way, new classes and old classes are balanced in a mini-batch. Compared with combining the exemplar data and new class samples, SL further alleviates the imbalance between the past and new classes. The details of our class-incremental method are introduced in Algorithm 1. To better understand the proposed algorithm ICAC, Fig. 4 shows the schematic of the incremental process from two to three classes in the training process.

Algorithm 1 ICAC Training Step

Input: $\mathcal{X} = (X_{s+1}, \ldots, X_{s+t}), \mathcal{P} = (P_1, \ldots, P_s)$ // new classes data, old exemplar sets
Input: $\Theta^s, \hat{\Theta}^s, A_s$ // current model, frozen current model, current anchored class centers
Output: Θ^{s+t} // model trained on t new classes
$\Theta^{s+t} \leftarrow \Theta^s$ // add output nodes for new classes
$A_{s+t} \leftarrow A_s$ // add new anchored class centers for new classes
for $(x, y) \epsilon\ \mathcal{X}$ **do**
  update for mini-batch data in $\mathcal{X}$
  $o = \Theta^{s+t}(x)$ // $o = \{o_{\mathrm{old}}, o_{\mathrm{new}}\}$.
  compute classification loss $\mathcal{L}_{\mathcal{X}}^{CAC}$ with o (Eq. 5)
  softmax over old class logits $\sigma(o_{\mathrm{old}})$
  compute distillation loss $\mathcal{L}_{\mathcal{X}}^{KD}$ (Eq. 9)
  load a mini-batch from exemplars set $(x', y') \epsilon\ \mathcal{P}$
  $o' = \Theta^{s+t}(x')$ // $o' = \{o'_{\mathrm{old}}, o'_{\mathrm{new}}\}$
  compute classification loss $\mathcal{L}_{\mathcal{P}}^{CAC}$ with o' (Eq. 5)
  softmax over old class logits $\sigma(o'_{\mathrm{old}})$
  compute distillation loss $\mathcal{L}_{\mathcal{P}}^{KD}$ (Eq. 10)
  $\mathcal{L} = (\mathcal{L}_{\mathcal{X}}^{CAC} + \mathcal{L}_{\mathcal{P}}^{CAC}) + \xi * (\mathcal{L}_{\mathcal{X}}^{KD} + \mathcal{L}_{\mathcal{P}}^{KD})$
end

In the test stage, the prototype vector for each training class is computed by the corresponding exemplars, and the input samples are classified by the nearest-mean-of-exemplars classification strategy as Algorithm 2. Figs. 5 and 6 show the t-SNE visualization of the test class representations in the feature space when learning MSTAR and OpenSARShip incrementally, respectively. It can be seen from the figure that ICAC can still commendably cluster the testing class samples and show intraclass compactness as well as interclass separability, although there are only 30 reserved samples in each increment process for each old class when training. Meanwhile, the prediction of the network will bias toward the new class when the number of classes is increasing. However, to a certain extent, ICAC eases this problem. For example, in Fig. 5(e) and (f), only a small number of other class samples overlap with the new class in the distribution.

Algorithm 2 ICAC Test Step

Input: x // image to be classified
Input: $\mathcal{P} = (P_1, \ldots, P_{s+t})$ // class exemplar sets
Input: $\Theta^{s+t} : \mathcal{X} \longrightarrow \mathbb{R}^{s+t}$ // feature map
for $y = 1, \ldots, s+t$ **do**
  $\mu_y \longleftarrow (1/|P_y|) \sum_{p \epsilon P_y} \Theta^{s+t}(p)$ // mean-of-exemplars
end for
$y^* \longleftarrow \arg\min_{y=1,\cdots,s+t} \|\Theta^{s+t}(x) - \mu_y\|$ // nearest prototype
Output: class label y^*

IV. Experimental Evaluation

A. Experimental Setup

1) Dataset: Over the years, there have been many approaches to SAR ATR, and most of them concern about the detection, classification, and identification of targets, scene, and context. In this article, the incremental learning for SAR ATR is concerned about. The MSTAR dataset and the OpenSARShip dataset are used for the experimental evaluation. The MSTAR dataset is organized under a Defense Advanced Research Projects Agency (DARPA) program, and it consists of the X-band and HH polarization SAR images with the 0.3-m resolution for multiple targets [37]. The MSTAR dataset contains ten types of targets: 2S1 (cannon), BMP2 (tank), BRDM2 (truck), BTR60 (armored car), BTR70 (armored car), D7 (bulldozer), T62 (tank), T72 (tank), ZIL131 (truck), and ZSU23/4 (cannon). These targets are captured at two different depression angles (15° and 17°) with 190–300 different aspect versions. The details of the MSTAR dataset are shown in Table I. In the experiments, the images with the depression angles of 17° and 15° are applied as the training set and the test set severally. In addition, the effectiveness of the proposed method is proven again through OpenSARShip [38]. OpenSARShip is ship data derived from the Sentinel-1 and contains four types of targets: bulk carrier, container ship, general cargo, and tanker. OpenSARShip is mainly based on civilian ship target samples, and the same type target samples often have morphological differences. In addition, the resolution of this dataset has decreased compared with MSTAR. In these experiments, 1239 ship slices have been cut out from the large scene image of OpenSARShip, and the details of the constructed ship dataset utilized in the experiment are shown in Table II. In the data preprocessing

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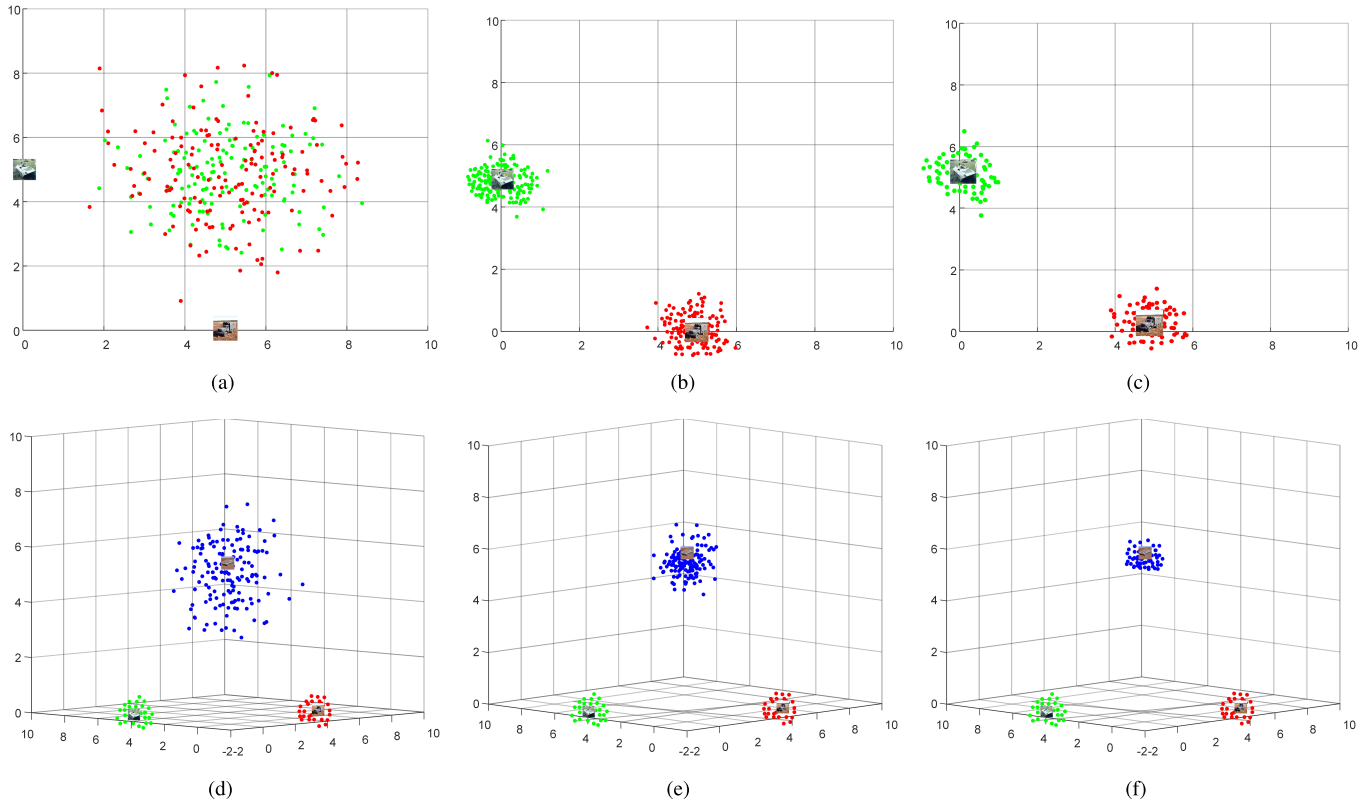


Fig. 4. Schematic of the incremental process from two to three classes. (a) Red and green dots represent the corresponding features of the two classes' samples, and the position of the image on the coordinate axis represents the position of the anchored class center. (b) Features of different classes are clustered around the corresponding anchored class centers after training. (c) Solid dots represent m samples selected as exemplars for each class. (d) Blue dots represent new class features. When a new class arrives, the feature space is increased from 2-D to 3-D, and a new anchored class center will be generated in the network to adapt to the new class. (e) Features of new class are all clustered around the new anchored class center. (f) Select m samples as exemplars for the new class.

TABLE I
SUMMARY OF THE USED MSTAR DATABASE

Target Type	Training Set (17°)	Testing set (15°)
2S1	299	274
BMP2	233	195
BRDM2	298	274
BTR60	256	195
BTR70	233	196
D7	299	274
T62	299	273
T72	232	196
ZIL131	299	274
ZSU23/4	299	274

stage, all images are resized to 64×64 but without any other preprocessing.

2) Evaluation Protocol: The evaluation protocol defined in [29] is adopted in this article, and a new protocol called average retention ratio (ARR) is proposed, which is about the ratio of the stored exemplars in the new class samples during each incremental step. For the MSTAR dataset, four random classes are as the initial classes, and the remaining six classes are as the incremental classes with the incremental steps of one, two, and three classes. For the OpenSARShip dataset, two random classes are as the initial classes, and the remaining two classes are as the incremental classes with the incremental steps of one class. After each incremental step, the resulting model is evaluated on the test data, which is composed of all the trained classes. Average incremental accuracy on the test set is used as the evaluation measure. Several different splits are evaluated with initial classes and incremental classes in each split, and then, the averages as well as the standard deviations of the results will be reported. To ensure that the results are comparable, the class order is identical for all evaluated methods.

TABLE II
SUMMARY OF THE USED OPENSARSHIP DATABASE

Target Type	Training Set	Testing set
Bulk carrier	204	100
Container ship	204	99
General cargo	241	98
Tanker	196	97

B. Class-Incremental Recognition Performance

In this section, six types of experiments are performed on the MSTAR dataset, and four types of experiments are

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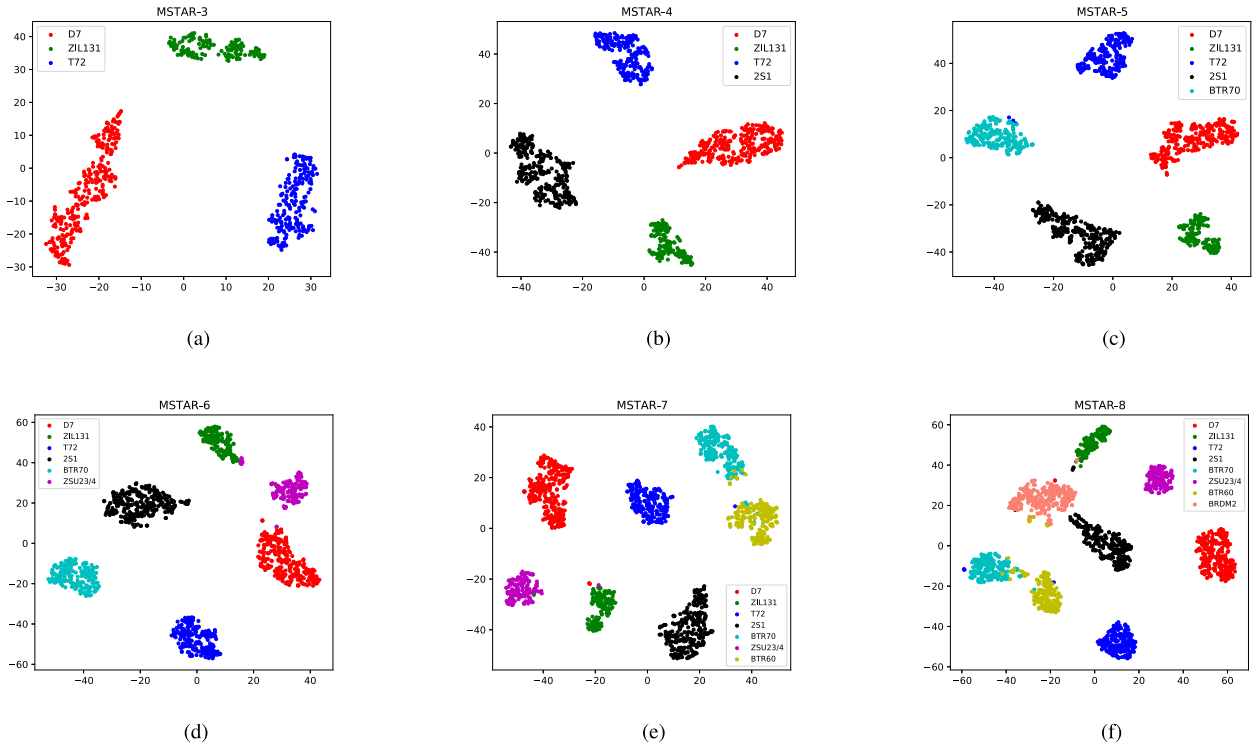


Fig. 5. t-SNE visualization of the test class representations in the feature space when learning MSTAR incrementally. The output features are 2-D, which is suitable for visualization. (a)–(f) Incremental process of the eight classes formed from the initial three classes. The experiment parameters are set as the following: the anchor magnitudes $\alpha = 5$, the exemplar set sizes $m = 30$, and the anchor loss weight $\lambda = 0.1$.

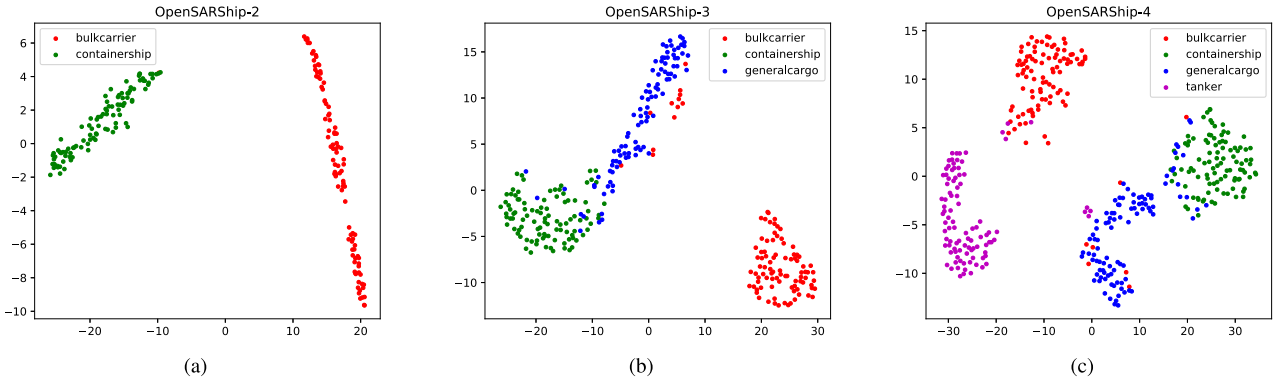


Fig. 6. t-SNE visualization of the test class representations in the feature space when learning OpenSARShip incrementally. The output features are 2-D, which is suitable for visualization. (a)–(c) Incremental process of the four classes formed from the initial two classes. The experiment parameters are set as the following: the anchor magnitudes $\alpha = 5$, the exemplar set sizes $m = 30$, and the anchor loss weight $\lambda = 0.1$.

performed on the OpenSARShip dataset. For the experiments on the MSTAR dataset, in the first one, an ablation study is performed to analyze the influence of different components of ICAC on the accuracy. Then, the average incremental accuracy of ICAC is compared with the other methods, such as compositional class-IL (CCIL) [27], iCaRL [29], joint training [39], end-to-end incremental learning (end-to-end) [40], and class boundary exemplar selection-based incremental learning (CbesIL) [6] in a class-IL scenario. The third experiment evaluates the method under the condition of different anchor magnitudes and compares the classification performance under four different anchor magnitudes: $\alpha = 1$, $\alpha = 5$, $\alpha = 10$, and $\alpha = 20$. Next, the method under the condition of different exemplar set sizes $m = 30$, $m = 40$, and $m = 50$ for each class is evaluated, respectively. Finally, to verify the effectiveness of the proposed method, the fifth and sixth experiments are designed. The former tests the effect of saving storage resources, and the latter verifies the ability of reducing training time. For the experiments on the OpenSARShip, the experiment settings are as the same as the previous four experiments of MSTAR. Since the number of saved exemplars is as the same as MSATR, the fifth and sixth experiments do not need to be performed on OpenSARShip again. In all experiments, T represents the number of classes added per incremental step.

For the feature extractor network, the network proposed in [1] is adopted, and the $fc(\cdot)$ layers are modified in the experiments. The configuration of the employed network

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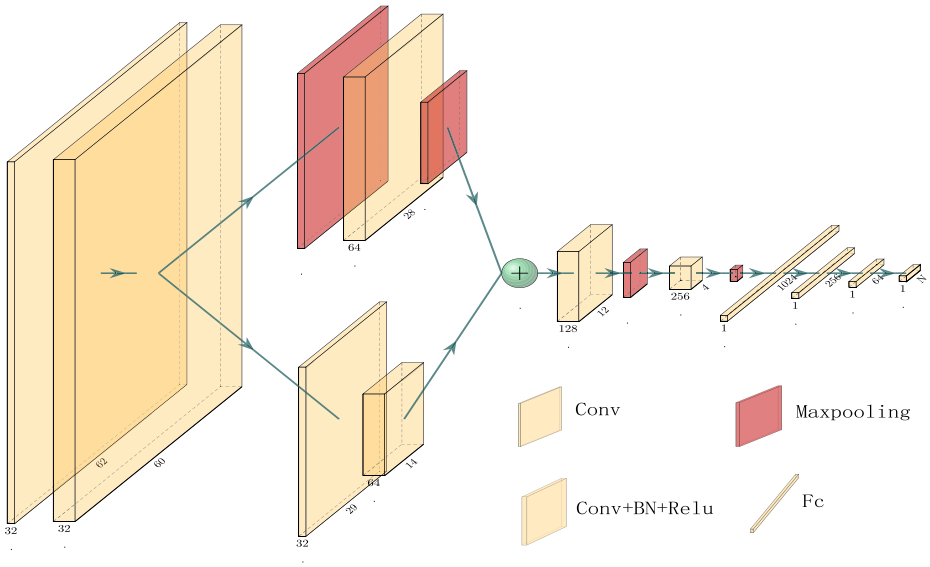


Fig. 7. Configuration of the employed network model. Conv represents convolution operation, BN represents batch normalization operation, and Fc represents full connection operation. N represents the number of classes to output.

TABLE III
MULTICLASS ACCURACY (AVERAGES ON SEVERAL DIFFERENT SPLITS) OF ICAC AND THREE HYBRIDS ON THE MSTAR DATASET

Model	T = 1							T = 2				T = 3		
	4	5	6	7	8	9	10	4	6	8	10	4	7	10
Hybrid1	99.45	96.51	94.36	93.31	92.65	89.68	87.87	99.45	95.23	92.45	90.70	99.45	94.81	92.44
Hybrid2	99.45	96.01	93.19	93.00	90.62	90.27	88.14	99.45	93.48	92.07	90.50	99.45	94.08	90.83
Hybrid3	98.47	95.94	92.95	91.82	86.90	86.51	85.19	98.47	94.40	91.67	88.02	98.47	94.33	86.99
ICAC	99.45	97.82	95.89	94.16	93.19	91.49	90.72	99.45	96.93	95.02	93.75	99.45	96.79	95.09

model is shown in Fig. 7. An adaptive moment estimation (Adam) optimizer is used with a learning rate of 0.0001 as well as a weight decay parameter of 0.00001 for ICAC, and then trains for 100 epochs with a batch size of 16. For all experiments in ICAC, the anchor loss weight λ is 0.1.

Due to the different incremental steps in MSTAR and OpenSARShip, the experiments on MSTAR are first introduced. For the first experiment, an ablation study is performed to analyze the influence of different components of ICAC on the accuracy. The main components used in this method include distillation loss, SL, and CAC with adaptively adding class anchor points. Three hybrid setups are created: hybrid1 does not use the distillation loss during training; hybrid2 uses neither the distillation loss, nor the SL (not using the SL means that the exemplar data and new class samples are combined in the training stage); hybrid3 does not use the CAC with adaptively adding class anchor points and use the standard CE loss instead. All the experiments make use of the exemplars and the exemplar set size $m = 30$.

Table III displays the results of the average incremental accuracies over all steps about ICAC and the three hybrids. The highest accuracy is marked in blue font. The comparison between ICAC and hybrid1 shows that knowledge distillation can alleviate catastrophic forgetting and improve the average incremental accuracy. By comparing hybrid1 and hybrid2, it can be found that, in most cases, using SL has a better performance than not using it. Finally, the comparison between the result of ICAC and hybrid3 clearly exhibits the effectiveness of the CAC method with adaptively adding new anchored class centers, which proves that ICAC can further alleviate catastrophic forgetting.

Table IV shows the class-incremental recognition performances of ICAC, CCIL, iCaRL, end-to-end, CBesIL, and joint training. In this experiment, the exemplar set size $m = 30$ is set for all the algorithms except joint training, and the anchor magnitude $\alpha = 5$ is also set for ICAC. CCIL is a method that combines knowledge distillation and exemplar retention. It proposes a CCIL model that isolates the underlying reasons for catastrophic forgetting in class-IL. ICaRL is introduced in [29], which uses the exemplars in combination with distillation to avoid catastrophic forgetting, and it learns strong classifiers as well as data representation simultaneously. End-to-end [40] presents a novel approach to train CNN in an incremental manner using a combination of CE and distillation loss functions. CbesIL [6] is a class boundary exemplar selection-based incremental learning method, which preserves the previous recognition capabilities in the form of the class boundary exemplars.

As it can be seen, ICAC achieves the best average incremental accuracy from the beginning to the end, except for the joint training as the upper boundary achieved by these methods. The highest accuracy is marked in blue font except the joint

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TABLE IV

MULTICLASS ACCURACY (AVERAGES ON SEVERAL DIFFERENT SPLITS) OF ICAC AND OTHER METHODS ON THE MSTAR DATASET

Model	T = 1							T = 2				T = 3		
	4	5	6	7	8	9	10	4	6	8	10	4	7	10
Joint-training	99.38	99.33	99.51	99.29	99.26	99.18	99.32	99.38	99.51	99.26	99.32	99.38	99.29	99.32
CCIL	98.47	91.04	88.78	83.33	82.08	75.85	74.84	98.47	91.41	87.13	81.90	98.47	92.40	84.16
iCaRL	94.47	91.61	87.52	86.45	83.29	82.51	80.02	94.47	90.16	87.20	84.12	94.52	88.46	84.74
End-to-End	98.67	96.63	89.76	81.03	76.62	73.09	71.20	98.00	91.84	80.55	72.87	98.00	89.49	73.34
CBesIL	98.06	82.80	77.44	76.33	75.11	73.13	72.15	98.06	81.13	76.39	70.21	98.06	77.49	71.75
ICAC	99.49	98.04	96.76	95.65	94.83	93.42	91.76	99.49	97.35	96.30	94.77	99.49	97.13	95.64

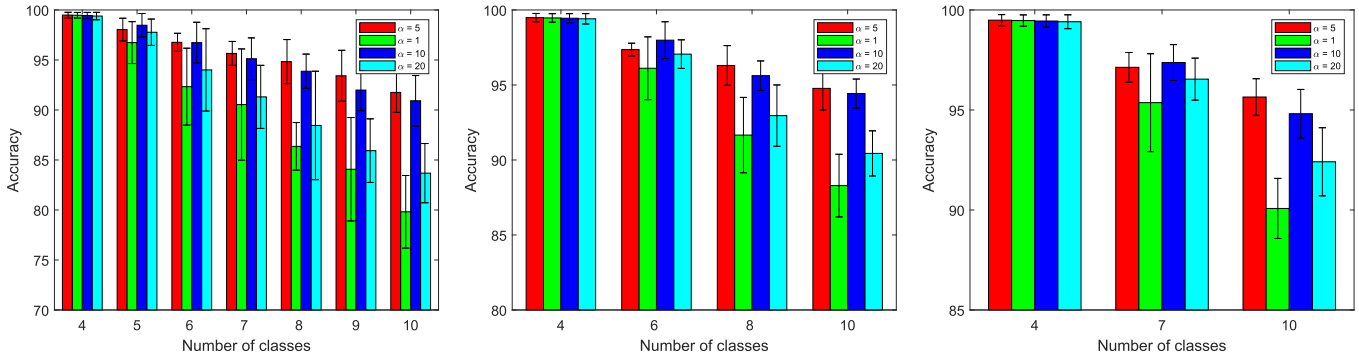


Fig. 8. Multiclass accuracy (averages and standard deviations on several different splits) on the MSTAR dataset under the condition of different anchor magnitudes α. (a) Add one class per incremental step, $T = 1$. (b) Add two classes per incremental step, $T = 2$. (c) Add three classes per incremental step, $T = 3$.

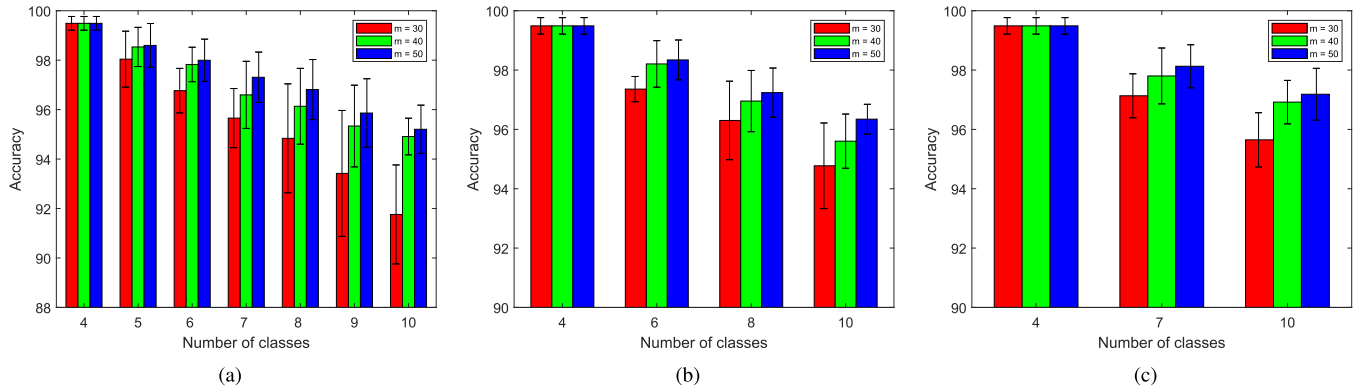


Fig. 9. Multiclass accuracy (averages and standard deviations on several different splits) on the MSTAR dataset under the condition of different exemplar set sizes m for each class. (a) Add one class per incremental step, $T = 1$. (b) Add two classes per incremental step, $T = 2$. (c) Add three classes per incremental step, $T = 3$.

training. The initial average accuracy of these algorithms is different because of the different classification strategies and training skills.

Fig. 8 illustrates the effects of different anchor magnitudes α, which represents the positions of the class anchor points on the axis. It can be observed that different anchor magnitudes have different effects on the average incremental accuracy. ICAC performs better when α is 5 and 10 and poorer when α is 1 and 20. The reason is mainly in two aspects; one is that α should not be set too large (e.g., 20). Otherwise, it will be difficult for the features of samples to gather around the anchor point, which means that the network training is difficult to converge, resulting in a decrease in identification accuracy. The other is that α should not be set too small (such as 1). If not, the distance between classes will become very small, which leads to the difficulty of distinguishing between classes and the decline of identification accuracy. Therefore, it is a relatively proper choice to set anchor magnitude between 5 and 10. Besides, the anchor magnitudes in other experiments are also set to 5.

Fig. 9 provides the effects of different exemplar set sizes for each class. Since incremental learning can only save a small number of exemplars, the exemplars number is set between 10% and 20% in the MSTAR dataset. So, the saved exemplars

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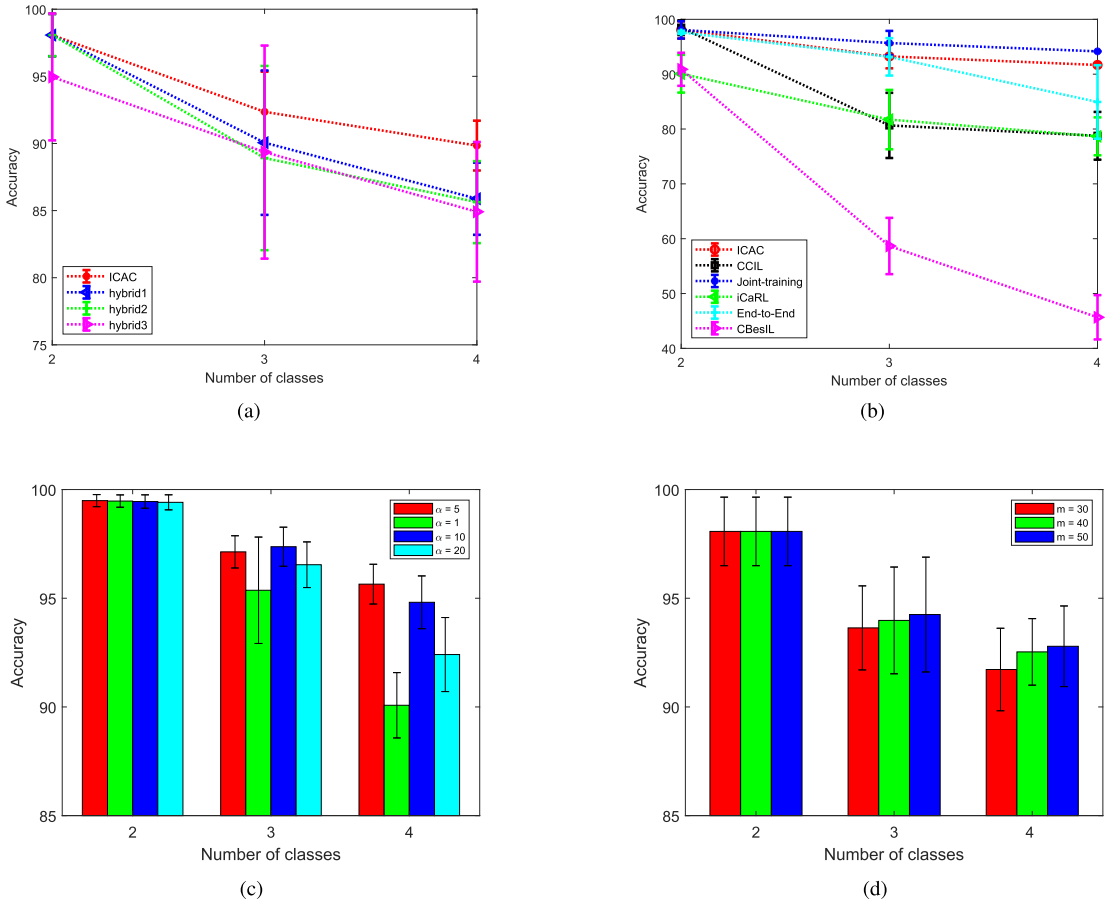


Fig. 10. Four types of experiments are performed on OpenSARShip dataset. (a) Multiclass accuracy of ICAC and three hybrids on the OpenSARShip dataset. (b) Multiclass accuracy of ICAC and other methods on the OpenSARShip dataset. (c) Multiclass accuracy on the OpenSARShip dataset under the condition of different anchor magnitudes α. (d) Multiclass accuracy on the OpenSARShip dataset under the condition of different exemplar set sizes m for each class.

number m is set to [30, 40, 50]. Obviously, the larger the exemplar set size m, the higher the average incremental accuracy.

To show that ICAC can reduce the required storage resources effectively, an evaluation protocol called retention ratio (RR) about the ratio of the exemplars stored in each incremental step to the new class samples is proposed. Since the six different splits with different initial four classes and different incremental six classes are evaluated, the ARR is applied instead of the RR. The definition of ARR is as the following:

$$\mathrm{ARR} = \frac{\mathrm{m}NT}{\sum_{k=1}^{N} N_{\mathrm{new}}^{k}} \tag{11}$$

where m is the exemplar set size, N is the number of experiments with different splits, and T is the incremental step value. N_{new} represents the total numbers of new class samples for each incremental step.

In the fifth experiment, the ARR under different exemplar set sizes m with different classes per incremental step is calculated. For $m = 30$, the ARR is from 0.1054 to 0.1178; for $m = 40$, the ARR is from 0.1406 to 0.1571; and for $m = 50$, the ARR is from 0.1757 to 0.1963. It is clear that ICAC only stores a small number of exemplars from the new class samples of each increment.

For the experiments on the OpenSARShip, four types of experiments are performed on the OpenSARShip dataset, as shown in Fig. 10. For the first experiment, an ablation study is performed to analyze the influence of different components of ICAC on the accuracy. As it can be seen from Fig. 10(a), compared with the ablation experiment on MSTAR, similar results are obtained, and the mutual comparison between the hybrids proved the effectiveness of each component. Then, the average incremental accuracy of ICAC is compared with the other five methods, which can be seen from Fig. 10(b). As we can seen, ICAC still achieved the highest average incremental accuracy compared with other algorithms. Fig. 10(c) provides the method under the condition of different anchor magnitudes and compares the classification performance under four different anchor magnitudes: $\alpha = 1$, $\alpha = 5$, $\alpha = 10$, and $\alpha = 20$. Similar to the experiment on MSTAR, ICAC performs better when $\alpha = 5$ is 5 and 10 and poorer when $\alpha = 5$ is 1 and 20. Finally, Fig. 10(d) exhibits the results of this method under the condition of different exemplar set sizes $m = 30$, $m = 40$, and $m = 50$ for each class, respectively. Also, the larger the exemplar set size m, the higher the average incremental accuracy.

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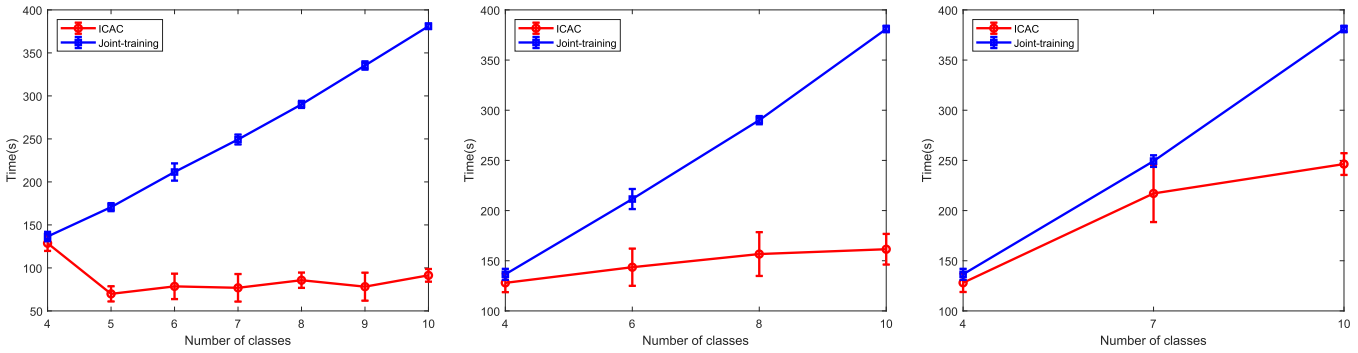


Fig. 11. Comparison of ICAC and joint training in training time on the MSTAR dataset. (a) Add one class per incremental step, $T = 1$. (b) Add two classes per incremental step, $T = 2$. (c) Add three classes per incremental step, $T = 3$.

Compared with joint training, incremental learning does not require repeated training; thus, it can save a lot of time. In the last experiment, a comparison of ICAC and joint training in training time is shown in Fig. 11. It can be seen that ICAC significantly reduces the training time compared with joint training, for any incremental steps. As the number of classes increases, the time required for joint training increases almost linearly, whereas the time required for ICAC increases slowly.

V. Conclusion

In this article, a new class incremental method called ICAC is proposed to solve the problems of class-IL for SAR ATR. To learn the new classes, ICAC adaptively adds new anchored class centers to minimize the intraclass distances and maximize the interclass distances, which can avoid the overlap between the distribution of the new and old classes. To solve the imbalance between old classes and new classes, SL is utilized among the exemplars. Experiments on the MSTAR dataset and the OpenSARShip dataset show that the method outperforms the current state-of-the-art incremental learning methods. In the experiment on the MSTAR dataset, the initial four classes are expanded to ten classes by incremental learning. In this process, only 30 samples (about one-tenth of the original training samples) are retained as exemplars for each old class. However, the recognition accuracy is only reduced by 10%, which is much lower than other incremental learning algorithms, proving that ICAC has better anti-forgetting ability compared with other methods. Experiments on OpenSARShip dataset confirm this conclusion as well.

This article verifies that the proposed ICAC is an efficient incremental learning frame of SAR ATR. ICAC is a method based on exemplars. In addition to using exemplars, the strong classification is mainly due to the use of three components: distillation loss, SL, and CAC with adaptively adding anchored class centers. These components are analyzed, and their impacts on the final performance are demonstrated.

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